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REVIEW

Electrorheological fluids: smart soft matter and characteristics

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An electrorheological fluid, a special type of suspension with controllable fluidity by an electric field, generally contains semiconducting or polarizable materials as electro-responsive parts. These materials align in the direction of the applied electric field to generate a solid-like phase in the suspension. These electro-responsive smart materials, including dielectric inorganics, semiconducting polymers and their hybrids, and polymer/inorganic composites, are reviewed in terms of their mechanism, rheological analysis and dielectric characteristics.

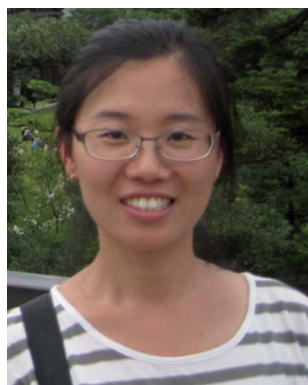
1. Introduction

“Smart” functional materials have attracted considerable attention because of their precisely controllable responses to their environments, including temperature, pH, light, electric or magnetic field, *etc.* The wide applications of these materials are attractive in a range of fields, such as sensing,^{1,2} drug delivery,³ electronics⁴ and vibration control.⁵ An electrorheological (ER) fluid^{6–12} is an active member in the family of smart materials, and is composed of electro-responsive soft materials and insulating carrier liquids. The suspended electro-responsive materials are mostly semiconducting or electrically polarizable particles that can be polarized in the presence of an external electric field to construct chain (and/or column)-like structures perpendicular to the parallel electrodes. Therefore, the rheological properties

including shear stress, dynamic modulus, and shear viscosity of the ER fluids can be changed substantially from liquid-like to solid-like. This controllable and reversible phase transition of ER fluids has potential applications in a range of engineering areas, such as microfluidics,^{13–15} damper system,^{16,17} tactile display¹⁸ and torque elements for use in rehabilitation devices.¹⁹ On the other hand, ER fluids not like their magnetic analogue magnetorheological fluids are not applied widely,^{20–22} due to technical problems of designing devices with applied high electric fields and the lack of suitable materials with high ER effects, good dispersibility, thermostability, *etc.* Therefore, a study of ER materials should include the ER characteristics and a search for suitable materials for applications.

ER materials cover a wide variety of materials ranging from inorganic to organic polymers and inorganic/organic composite materials with particle sizes in the nanometer to micrometer scale and various particle shapes including spherical and non-spherical (*e.g.*, rod-like, fibrous, and sheet).^{23,24} In addition to the materials

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themselves, the above-mentioned factors significantly influence the ER effect as well as the stability of the corresponding ER fluid. In the early stages of ER fluid research, extrinsically polarizable particles, such as silica, aluminosilicate and cellulose, which contain a range of additives to induce polarizability, have attracted enormous attention.¹¹ However, these particles, which were used as hydrous ER fluids, have some drawbacks in their applications including device corrosion, water evaporation, narrow operation temperature range and dispersion instability. An additional study of ER materials focused on anhydrous inorganic TiO₂ and perovskites with or without modification by metallic cations or polar molecules.²⁵ On the other hand, a series of conducting polymers with controlled and relatively low conductivity (10⁻⁶ to 10⁻¹⁰ S cm⁻¹) are encompassed in anhydrous ER materials, and have attracted considerable attention. Polyaniline (PANI), polypyrrole (PPy) and their derivatives are popular candidates for these kinds of ER materials. Their composites with inorganics or polymers are under intense study as ER materials owing to their enhanced chemical, physical characteristics compared to individual pure materials. The design of core-shell particles containing conducting polymers or other inorganics as the electro-responsive part has attracted considerable attention for their interesting ER effects.^{26–28}

This paper reviews the soft matter materials employed in ER fluids including their mechanisms, synthesis methods, characteristics, and ER effects observed using a range of rheological measurement methods. The main focus is in general to provide insight into the material rheology aspects of electrorheology rather than the physical mechanisms and dynamic analysis. On the other hand, to improve the understanding of ER materials, the basic introduction is started from the mechanisms to which numerous physicists have made great contributions.^{29,30} Note that although recently published detailed mechanisms can be found,¹¹ there are few overall reviews of the various ER materials covering both inorganic and organic objects, their characteristics and ER properties.²⁴

2. ER mechanisms

2.1 Mechanisms for conventional ER fluid

The mechanisms of the ER effect have been studied intensively. Winslow⁶ proposed that the ER effect was induced by the fibrillation of wet particles with a proper net charge in the insulating carrier liquid. The adsorbed water or surfactant in the particles was also considered necessary by many researchers,³¹ forming an adhesive water bridge between particles. An electric double layer model was also introduced based on wet ER particles.³² However, these mechanisms have become limited for anhydrous ER fluids.³³

The polarization model was then proposed to explain anhydrous ER materials, such that dielectric constant mismatch between the particles and medium is essential to the inter-particle force (F), in addition to the power law dependence (order of 2.0) of the yield stress on the electric field strength.^{29,34} Using an idealized ER fluid model with hard dielectric spheres (diameter: $2r$; real dielectric constant: ϵ_p) suspended in a Newtonian fluid (real dielectric constant: ϵ_m),³⁵ F can be expressed as follows:

$$F = k\epsilon_m(2r)^2f^2E^2S \quad (1)$$

where k is a constant, $f = (g - 1)/(g + 2)$ is the dielectric mismatch parameter and $g = \epsilon_p/\epsilon_m$. E is the applied electric field and S is a factor related to the particle microstructure. Eqn (1) suggests that a high particle dielectric constant would have a strong ER effect.

The conduction model^{36,37} is another highly recognized mechanism for an ER fluid, considering that the ER effect in a dc or a low frequency ac field was induced by mismatch of the conductivity of the particles and fluid medium. This model can predict the current density and field dependence of yield stress on the electric field for ER fluids with a highly concentrated particle volume fraction and high electric field strength. On the other hand, the model can be only used in situations where the particle microstructure was fully formed in an ER fluid.

Parthasarathy and Klingenberg²⁹ published an invaluable review of the ER mechanisms and microscopic models in the 1990s, covering all the physical mechanisms and models mentioned above, highlighting the significance of the polarization model in analyzing the particle interactions and macroscopic behavior of an ER fluid. Microscopic analytical theories that describe the structure and dynamics of an ER fluid under steady and oscillatory flow are also under study and several valuable models were introduced including the elliptical droplet model proposed by Halsey *et al.*^{38,39} and the chain model proposed by Martin *et al.*,^{40,41} based on the balance between the bulk and surface electrostatic energy, and the balance between the hydrodynamic and electrostatic forces, respectively. In addition, Tao and Sun⁸ predicted the ground state for ER fluids to be a body centered tetragonal (BCT) structure by treating the ER fluid as a suspension of point dipoles in a dielectric fluid generating the minimized dipolar energy when the dipoles are joined together by an electric field. This was also applied by other researchers to analyze the ER effect in real space. By considering the BCT structure as a ground state, Dassanayake *et al.*⁴² analyzed the structural transition in the ER fluid of fluorescent silica hard spheres from the nonequilibrium structures of one-dimensional “chains” and two-dimensional “sheets” to BCT crystals according to the particle concentration,⁴³ whereas fluorescent poly(methyl methacrylate) (PMMA) soft spheres were observed to perform differently from silica hard spheres with the appearance of space-filling tetragonal crystals by increasing the concentration of spheres.⁴⁴ In this model system of a monodisperse colloidal suspension, a real-space study was performed carefully on crystal melting by tuning the repulsive and anisotropy interactions by varying the salt concentration and applied electric field.⁴⁵ This suggests that investigations of the modeling mechanisms of ER fluids were developed by confirming the modeling crystal structures in real space using a colloidal suspension of hard spheres.

2.2 Mechanisms for giant ER fluid

The concept of a giant electrorheological (GER) effect was first introduced by Wen *et al.*⁴⁶ They examined nanoparticles of a urea-coated BaTiO(C₂O₄)₂-based ER fluid, which could reach a yield stress orders of magnitude larger than the theoretical upper bound of the static yield stress in a conventional ER fluid. The static yield stress of the GER fluid showed a linear relationship

with the external applied electric field and varied by $1/R$ (R , radius of the GER particle), suggesting that smaller particles would generate a larger effect. The GER effect was only discovered in silicone oil according to the wetting effect between the nanoparticles and continuous phase. The discovery of GER fluid expanded the possibility of practical applications of an ER fluid because the low yield behavior of conventional ER fluids is considered one of the main obstacles to its applications. To understand the GER effect, Lu *et al.*^{47–49} suggested that polar molecules (*e.g.*, urea) adsorbed on the particles played key role in generating the GER effect. In this model, the attractive force between the particles consisted of a dipole–dipole interaction between the oriented polar molecules and the dipole–charge interaction between the polar molecules and nanoparticles (Fig. 1). Moreover, Sheng and Wen⁵⁰ recently reported a deep review of the mechanisms and dynamics in giant ER fluids compared to that of the dielectric ER fluids, where it was reiterated that the electric field could induce the alignment of urea dipolar filaments bridging the two boundaries of the nanoscale confinement using a molecular dynamics simulation.⁵¹ Fig. 2 shows that in the urea–silicone mixture, the urea molecular filaments are snaking through the silicone oil chains over the gap of two dielectric particles with aligned urea dipole molecules. Nonetheless, further study will be necessary to obtain a clearer understanding of the GER effect both theoretically and experimentally. All studies of the ER mechanisms and dynamic models should be verified using other materials, which is one of the reasons why material studies are being carried out continuously in the ER community.

3. Development and diversification of ER materials

Typical ER fluids are suspensions of 0.1–50 μm particles with volume fractions of 0.05–0.50. The first ER materials introduced were wet (*e.g.*, colloidal SiO_2 , aluminosilicate, starch, TiO_2 , cellulose, *etc.*), and the amount of water adsorbed on the surface of the ER particles had a significant effect on the ER effect.^{52–55} The fluids containing these moisture particles have many

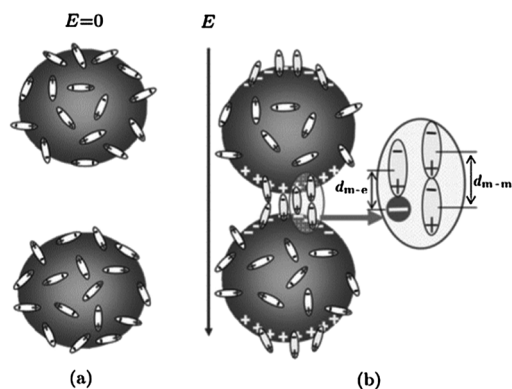


Fig. 1 Schematic diagram of the polar molecules aligned in the gap of the particles under an applied electric field. (a) Polar molecules adsorbed on the particle surface are oriented arbitrarily when $E = 0$. (b) The polar molecules in the gap of the particles are aligned along the field when E is sufficiently high. In the enlarged image (right), two types of interactions are shown: dipole–dipole interaction and dipole–charge interaction.⁴⁷

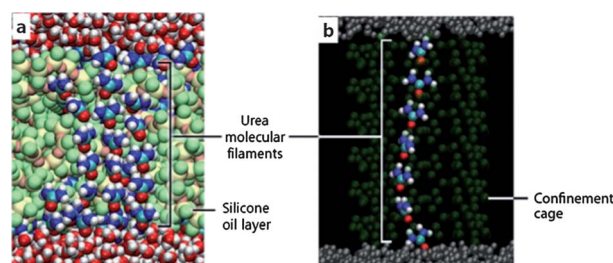


Fig. 2 (a) Urea molecular filaments across a silicone oil layer under a 0.2 V nm^{-1} electric field and (b) in a $\sim 1 \text{ nm}$ cylindrical confinement cage under a 0.1 V nm^{-1} electric field.^{50,51}

disadvantages that impede the practical applications of ER fluids.⁵⁶ Since the development of anhydrous ER fluids,^{57,58} they have been studied more intensively owing to their enhanced stability and potential applications. The materials developed since then involve zeolites, carbonaceous materials transformed from coal,^{59,60} conducting polymers, biopolymers, liquid crystals, and other dielectric inorganics. On the other hand, only the easily purchased inorganic zeolite or biomaterials (*e.g.*, starch) and the easily synthesized PANI particles are applied frequently in the study of ER mechanisms and models. Therefore, it is important to develop synthesis methods with mass production not only for a study of their mechanism but also for application research. The hydrous ER system is not covered in this review.

3.1 Porous inorganics

Zeolite, an aluminosilicate, is a frequently applied ER material because of its easy accessibility and high ER effect. The material was first reported to be a hydrous ER material but was later developed to anhydrous ER materials. The main zeolite formula was $M_{2/n}\text{OAl}_2\text{O}_3 \cdot x\text{SiO}_2 \cdot y\text{H}_2\text{O}$, where M is defined as the compensating metal cation with valence n . The metal cation is mobile under the stimulation of an electric field, which makes a critical contribution to the ER effect, even though there was no moisture in the particle.⁵³ The principal synthetic zeolites in commercial use are Linde Type-A (LTA) and Linde Type-X. For ER applications, 3A, 4A, 5A, 13X and even Y-Type zeolite have also been investigated,^{61–64} showing different ER effects owing to their different structural components and pore size in their frameworks. Tian *et al.*⁶³ reported a zeolite Y-based ER fluid with high shear yield stress ($>20 \text{ kPa}$) in a 5 kV mm^{-1} dc field by introducing a post-microwave treatment. This was the best result at that time. Bose *et al.*⁶⁵ also examined the effect of many other parameters including the zeolite content, particle size, cation composition and oil viscosity on the rheological properties using A-type zeolite. In addition, amorphous zeolite showed a larger ER effect than the crystalline material due to the presence of more impurities contributing to polarization.⁶⁶ Another mesoporous silica, MCM-41, was reported to show ER performance when dispersed in silicone oil owing to its large surface area and pores with active charge carriers and adsorbed moisture.^{53,67} An increase in shear stress with increasing electric field strength was observed in an ER fluid containing 20 wt% MCM-41. On the other hand, the yield stress was not as high as that displayed by the zeolite ER fluid. The large surface area and pore size on the

nanoscale enabled broad applications of MCM-41, not only as an electro-responsive material, but also in catalysis, separation, adsorption and advanced materials.⁶⁸

3.2 Conducting polymers

A range of conducting polymers with π conjugated structures, such as PANI,^{69–72} PPy,^{73–75} poly(*p*-phenylene) (PPP),^{76–78} poly-(acene quinone) radicals,^{32,79} poly(phenylenediamine),⁸⁰ oxidized poly(acrylonitrile), and polythiophene,^{81,82} in their appropriate state (doped or dedoped, reduced or oxidized) have been applied as ER candidates with suitable conductivity (10^{-6} to 10^{-10} S cm^{-1}) for ER measurements. Over this range (10^{-6} to 10^{-10} S cm^{-1}), a higher conductivity can result in current leakage and lower conductivity makes it difficult to generate an ER effect. Fig. 3 shows the particulate morphology of several conducting polymers.⁸³ In contrast to inorganic materials, polymeric ER materials are soft and non-abrasive to ER devices. Moreover, the lower density of the polymer leads to ER fluids with better suspension stability. Lee *et al.*⁷⁰ reported a surfactant-added PANI ER system referring to the surfactant effect on the stability and ER properties of the ER fluid.

Among the range of conducting polymers, PANI is the most popular because its good thermal stability, controllable conductivity by a doping and dedoping process, facile synthesis and relatively low cost, *etc.* PANI can be prepared by a chemical oxidation method with rapid mixing or stable interfacial reaction according to the desired morphology of PANI particles,⁸⁴ in which an aniline monomer is oxidized by ammonium persulfate (APS) in an acidic medium at low temperatures. High conducting PANI can be obtained in this process but pH adjustment with a basic solution of NaOH or NH_4OH is needed to reduce the conductivity of PANI. The dedoping level (pH value and time) of PANI is closely related to the current density of the ER fluid.⁷⁰ On the other hand, for a conducting polymer ER material, several problems still need to be solved, of which the high current density is most important. Many approaches have been proposed, *e.g.* introducing derivatives and copolymers of PANI,

such as poly(*o*-anisidine),⁸⁵ poly(aniline-*co*-*o*-ethoxy aniline)⁸⁶ with reduced conductivity, encapsulating the PANI particulates with an insulating layer,^{87,88} and introducing inorganic particles to the polymer matrix. A series of studies on the last approach have been published.^{89,90} Polyaniline/ Na^+ -montmorillonite (PANI/MMT) nanocomposite⁹¹ is an attractive product showing a considerably reduced current density compared to a commercial ER fluid.

3.3 Conducting material incorporated composites

3.3.1 Conducting polymer inserted inorganics. Inorganic nanoparticles with special morphologies, such as layered silicate (clay) and mesoporous silica (SBA-15, MCM-41), were applied as ER materials because of their special morphology on ER fluids involving the suspension stability and ER performance. On the other hand, these materials sometimes cannot be used directly because they exhibit a low ER effect compared to traditional inorganic materials when used directly.^{67,92} To overcome the disadvantages of these materials, conducting polymers were introduced to the interlayers of clay and mesopores of silica to prepare conducting polymer/inorganic composites.

Conducting polymer-intercalated Na^+ -montmorillonite (Na^+ -MMT)^{93–98} or organoclay (OMMT)⁹⁹ nanocomposites, *e.g.* PANI/clay or PPy/clay, and clay nanolayer-wrapped PANI spheres, have been evaluated as ER materials that exhibit novel chemical and physical stability compared to pure conducting polymers. A typical emulsion or microemulsion polymerization process has been introduced to prepare PANI/clay nanocomposites. In emulsion systems, pristine Na^+ -MMT, which has high swelling ability and sufficiently large interlayer spacing in water allowing the penetration of monomers of organic species, was applied as the clay component, and was dispersed in an aqueous medium for the emulsion (o/w) system. Surfactants, such as dodecylbenzene sulfonic acid (DBSA) and sodium lauryl sulfate (SDS), were critical elements contributing to the layer swelling of clay to introduce aniline monomers and acted as a dopant in the case of DBSA.^{100,101}

In addition to PANI, PPy is another well-known conducting polymer, and has been used as an ER material.⁷⁴ PPy-intercalated clay nanocomposites, such as PPy/ Na^+ -MMT, have been synthesized using a range of methods including solvent casting¹⁰² and inverted emulsion.⁹⁷ In the solvent casting process, highly soluble PPy was synthesized first by *in situ* polymerization using sodium di-(2-ethylhexyl sulfosuccinate) as a dopant. The PPy/clay nanocomposite was then prepared in chloroform, which contained well exfoliated OMMT (Cloisite 25A). Fig. 4 shows

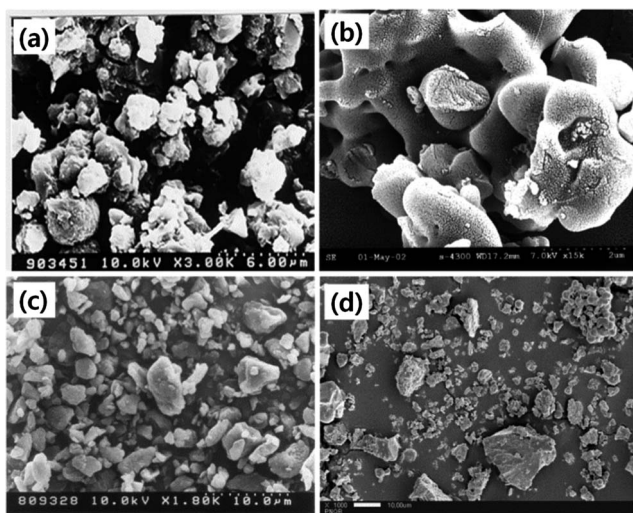


Fig. 3 SEM images of several particulate conducting polymers on a micron scale: (a) PANI, (b) PPy, (c) PPP, (d) PNQR.⁸³

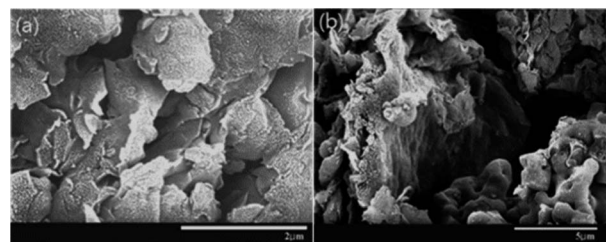


Fig. 4 SEM images of pure OMMT (a) and after intercalation by PPy via solvent casting (b).¹⁰²

the morphological properties of both OMMT (a) and its intercalated nanocomposite with PPy (b). Fig. 4(b) shows both granular PPy and plate-like clay, indicating that the dissolved PPy is not only intercalated into interlayers but also adsorbed onto the surface of the clay layers.

For mesoporous MCM-41 or SBA-15-contained ER composites, either PANI or PPy polymer chains were synthesized within the channels of their mesopores. Before polymerization, the corresponding monomer was first introduced to the channels of dry powders of MCM-41 or SBA-15 in a vacuum and then immersed in a HCl solution followed by the addition of an oxidant initiator, APS. Fig. 5 presents TEM images of MCM-41 before (a) and after (b) the polymerization of pyrrole.¹⁰³ Pure mesoporous MCM-41 is composed of well-ordered hexagonal channels that become smeared and dark after the insertion of PPy. Measurements of the N₂ adsorption/desorption isotherm is an effective way of confirming the successful location of the polymers within the channels. Fig. 6 shows the N₂ adsorption/desorption isotherm curves of MCM-41 and PANI/MCM-41. The residual pore volume of PANI/MCM-41 was reduced from 0.96 to 0.63 cm³ g⁻¹ due to pore filling by PANI.¹⁰⁴ Similar methods were used to prepare PANI- or PPy-filled SBA-15 composite particles.^{105–107} General conducting polymer/inorganic composites containing TiO₂ or BaTiO₃ nanoparticles were also synthesized and applied because of their unique dielectric properties.^{108–110}

3.3.2 Core-shell-structured ER particles. The conducting polymers used as ER materials have many different forms and can be inserted or intercalated into inorganic particles. On the other hand, coating or encapsulation on the template particles is another way of using conducting materials in ER fluids, which is not limited to the polymers. Thereby, core-shell-structured ER particles containing conducting materials as either core or shell species have become a hot topic recently with advantages, such as enhanced ER effects, diverse ER mechanisms, lower particulate density, and lower cost than using conducting materials only.

Core-shell structured particles can be prepared using a two-step procedure including the synthesis of the polymer core microspheres and the coating process by conducting materials as electro-responsive shells. A range of polymerization methods have been used to prepare monodisperse polymer core spheres, such as dispersion polymerization, traditional emulsion or seeded emulsion polymerization depending on the particle size or expected shape.^{111–113} Special techniques are then needed to direct the coating process on core particles. Surfactants or grafting

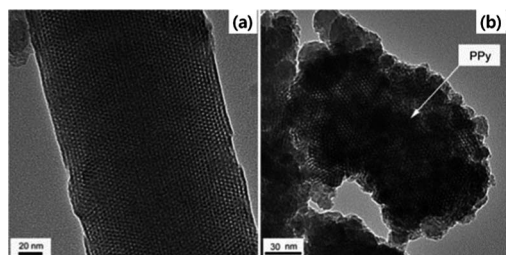


Fig. 5 TEM images of MCM-41 (a) and PPy/MCM-41.¹⁰³

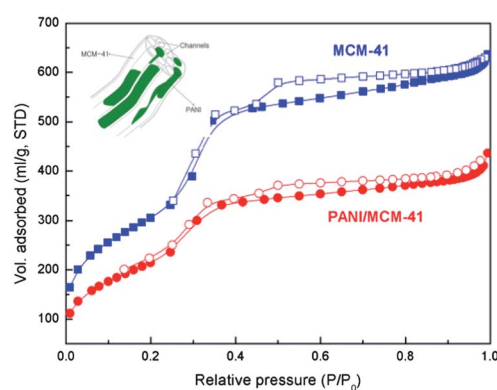


Fig. 6 N₂-adsorption isotherm curves (solid: adsorption; open: desorption) of pure MCM-41 (square) and PANI-inserted MCM-41 (circle).¹⁰⁴

reagents are sometimes necessary to modify the raw polymer cores, and allow easy chemical interactions or physical adsorption with conducting shell materials. When the core particles and coated materials are attracted to each other by some physical interactions, e.g., π - π stacking or hydrogen bonding, a conducting layer of CNT, graphene oxide (GO) or the conducting polymers, such as PANI and poly(3,4-ethylenedioxythiophene)/poly(styrene sulfonate) (PEDOT/PSS) could be adsorbed directly on the core particles.

For the coating technique, particularly on the surface of polymeric microspheres, the surfactant is extremely useful as an intermediary material because of its special structure with the ability to connect hydrophilic and hydrophobic materials. Jin *et al.*¹¹⁴ prepared fine microparticles of polystyrene (PS) and PMMA with a CNT coating using a range of surfactants including anionic sodium dodecyl sulfate and sodium dodecylbenzene sulfonate, cationic cetyl trimethylammonium bromide (CTAB), and nonionic Triton X-100. The application of a surfactant was also helpful in dispersing the CNTs in an aqueous medium by sonication. Nevertheless, the surfactants had different coating effects, with the CTAB showing the best result. Recently, CNT-coated snowman-like particles were produced in a similar manner in an aqueous solution of CTAB containing well dispersed CNTs.¹¹⁵ Fig. 7(a) shows SEM images of CNT coated PS particles. Well adsorbed CNTs were observed either to be dense or thin due to the chemical properties of the particle surface.

A surfactant can be used to attract the monomers of conducting polymers onto the surface of polymer spheres. The PMMA/PANI core-shell particles were prepared using the surfactant, DBSA,^{116,117} which also acts as a dopant for PANI. Following a similar method, conducting PANI-coated anisotropic snowman-like particles were prepared from a simple process,¹¹⁸ in which snowman-like particles were pre-fabricated using a seeded emulsion polymerization method with cross-linked PMMA micro-seeds.^{113,119} The snowman-like particle is also an anisotropic ER material, which includes particles with rod-like and fibrous shapes. The ER fluids of anisotropic particles showed improved ER performance and lower off-state viscosity than the ER fluids of spherical particles. Yin and Zhao published a detailed and interesting review on this topic.²⁴

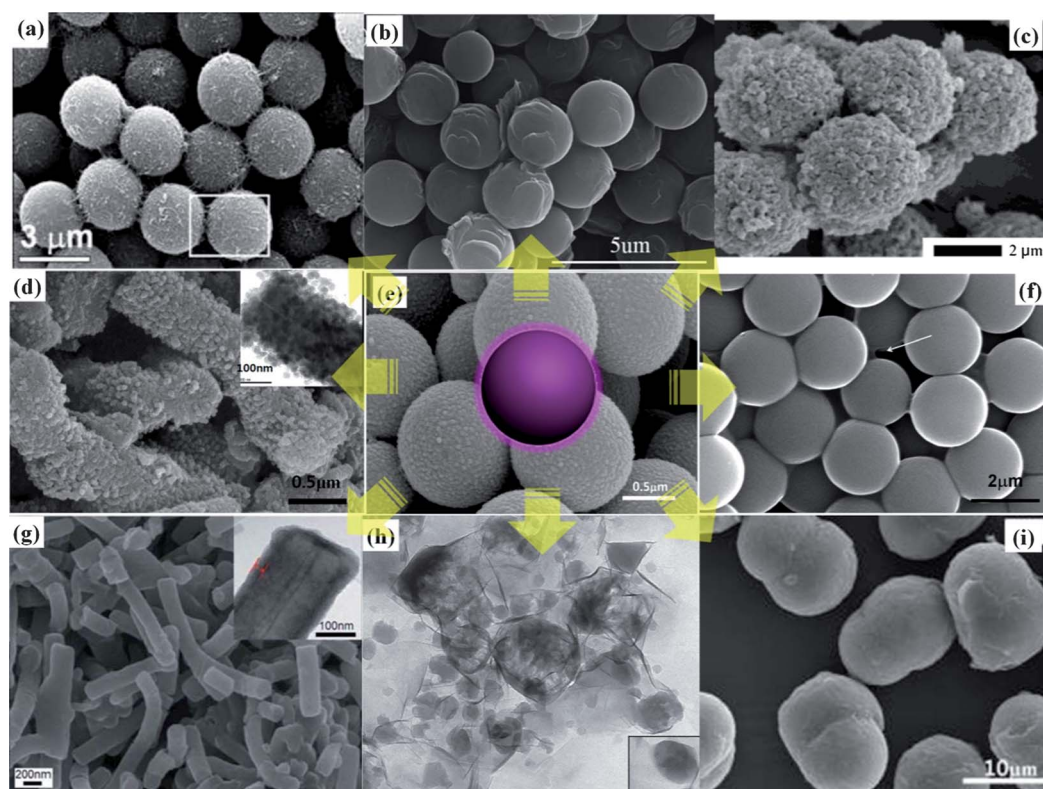


Fig. 7 SEM and TEM images of various core-shell ER particles with incorporated conducting materials: (a) PS/CNT,¹¹⁴ (b) PS/GO,¹²⁴ (c) PMMA/PANI,¹¹⁶ (d) PANI/SiO₂,¹³⁰ (e) PS/PANI,¹²⁷ (f) PS/PEDOT/PSS,¹²⁶ (g) PANI/TiO₂,¹²⁹ (h) PANI/Clay,¹³¹ (i) snowman-like PMMA/PANI.¹¹⁸

To enhance the adhesive interaction between the inner core and outer shell in core-shell particles, a chemical interaction was introduced by applying a grafting agent to the core particles. Uniform PANI-coated PMMA particles were designed for this purpose.¹²⁰ The PMMA particles were coated with epoxy groups by swelling the particles with glycidyl methacrylate (GMA), and reacted with oxydianiline, which can connect to the PMMA particles *via* an amine-epoxy reaction from one side and start the oxidation of aniline from the other side. This process was recently improved by adding another cross-linker, ethylene glycol dimethacrylate (EGDMA), to the swelling process. On the other hand, Kim *et al.*¹²¹ introduced 3-aminopropyltrimethoxysilane-anchored PGMA particles with densely coated carboxylic acid-functionalized CNTs through a covalent interaction.

Other simple ways were also recently developed to prepare core-shell ER particles, *e.g.* *via* hydrophobic interactions. The GO with lower conductivity and good processability in both aqueous and organic solvents compared to graphene, was lately reported to be suitable for ER materials when coated on polymer spheres or blended with either PANI or inorganics.^{122–125} Core-shell particles of PS/GO were fabricated in a facile manner, *i.e.* the PS particles and GO were dispersed in ethanol by sonication and then mixed at higher temperatures with constant stirring.¹²⁴ The well dispersed GO sheets were attached to the PS microspheres by strong π - π stacking interaction. Using a similar method, PEDOT/PSS-coated PS particles were also prepared with a thin shell thickness of approximately 5 nm, as determined by TGA.¹²⁶ The controllable (swelling-releasing) synthesis of PANI on the PS particle surface was also realized using aniline

monomer-swelled PS spheres due to the hydrophobic affinity between aniline and PS chains.¹²⁷ Particles coated uniformly with conducting polymers could be prepared more easily in this manner than the method reported previously.

Novel core-shell ER particles containing a conducting core and insulating shell were prepared, and found to have a lower current density in an applied electric field. An insulating coating process was applied to the PANI nanofibers because fibrous materials were found to have a higher ER effect than spherical or rod-like particles.²³ Coaxial cable-like PANI/TiO₂ nanofibers^{128,129} prepared *via in situ* precipitation were reported to exhibit slightly lower yield stress, but significantly lower current densities. On the other hand, PANI fibers coated silica-nanoparticles with exhibited a higher ER effect than the pure fibers due to the roughly adhered silica particles.¹³⁰ A novel emulsion method, “Pickering” emulsion^{99,131} was developed to prepare conducting polymer microspheres coated with solid particles. The applied solid particles could be clay, silica, ZnO or even graphene oxide. The synthesized particles would possess a lower conductivity and higher thermostability than bare conducting polymers. Fig. 7 shows SEM images of the aforementioned core-shell particles prepared using a range of methods.

3.4 Other organics or polymers

In addition to conducting polymers, polymers with polar groups, such as amino (–NH₂), hydroxyl (–OH) and amino-cyan (–NHCN), were also used as dry base ER materials. Among these materials, chitosan,^{132,133} a linear polysaccharide produced

by the *N*-deacetylation of chitin, and its derivatives have been studied extensively because of its good dispersibility in silicone oil and easy modification of the ER activity by functional groups. The deacetylation of chitin was performed in a NaOH aqueous solution with heating. The degree of deacetylation (DD)¹³⁴ had an attractive effect on the ER effect:¹³⁵ the shear stress of the chitosan-based ER fluid increased with increasing DD until a limit DD value was reached, above which there was an inverse trend. Other bio-polymers, such as starch and cellulose, showed an ER effect owing to their hydroxyl groups. On the other hand, large particles of these materials settled easily, resulting in poor sedimentation stability in their ER fluid.

Liquid crystal polymer (LCP) is another type of polymeric ER material. Both side-chain and main-chain LCPs could show an ER effect by themselves or upon dissolution in another liquid medium (e.g. the nematic solvent of 4'-(pentyloxy)-4-cyanobiphenyl or 4'-pentyl-4-cyanobiphenyl),^{136–138} forming miscible homogeneous ER fluids. These LCPs include polysiloxanes or general polymers with side chains of 4-((undecylenyloxy)benzoyl)-4'-methoxyphenyl, benzoic acid cyanophenylester with different spacer lengths between the backbone and mesogenic groups,^{139–142} poly(*n*-hexyl isocyanate) (PHIC), poly(6-(4'-cyanobiphenyl-4-yloxy)hexylacrylate),¹⁴³ and poly(γ -benzyl-L-glutamate).¹³⁶ The alignment of the LC domains could not generate an ER effect as high as the other inorganic or polymeric ER materials. Heterogeneous systems were developed by adding ionic or electronic powders to the LCP matrix. The maximum transmitted stresses of these systems reached 2.5 kPa at 1.6 kV mm⁻¹ using pyrolyzed polyacrylonitrile dispersed in PHIC.¹⁴⁴ The structural changes in the immiscible blend of a LCP and polyisobutylene (equal viscosity with the LCP) were examined quantitatively to determine the relationship between the structure and shear stress in the transient process.¹⁴⁵

3.5 High dielectric materials

The achievable polarizability, generated either from dielectric or conductive mismatch between the dispersed particles and carrier medium, is considered the dominant factor in the ER effect.¹⁴⁶ Therefore, a high dielectric constant (permittivity) has a positive influence on the ER effect. Generally, the permittivity is frequency dependent and used mostly in its complex form involving permittivity (ϵ') and loss factor (ϵ''): $\epsilon^* = \epsilon' + i\epsilon''$. The polarizability is related to the difference in permittivity ($\Delta\epsilon = \epsilon_0 - \epsilon_\infty$) and relaxation time ($\lambda = 1/2\pi f_{\max}$), which are obtained by fitting the dielectric spectra with the Cole–Cole formula.¹⁴⁷

$$\epsilon^* = \epsilon' + i\epsilon'' = \epsilon_\infty + \frac{\Delta\epsilon}{(1 + i\omega\lambda)^{1-\alpha}} \quad (2)$$

where ω is the angular frequency and $(1 - \alpha)$ represents the distribution of relaxation times over the entire frequency range. Generally, a strong ER effect is correlated with a large $\Delta\epsilon$ and a dielectric loss peak located in the range of 10^2 to 10^5 Hz.¹⁴⁸ Dielectric analysis is not only suitable for ER materials with a high dielectric constant, but is also important to conducting polymer-based ER fluids. Fig. 8 shows the dielectric spectra of an ER fluid system containing pure PANI fiber (PANIF) and silica-modified PANIF measured using a LCR meter and the Cole–Cole fitting plots using the parameters listed in Table 1. The

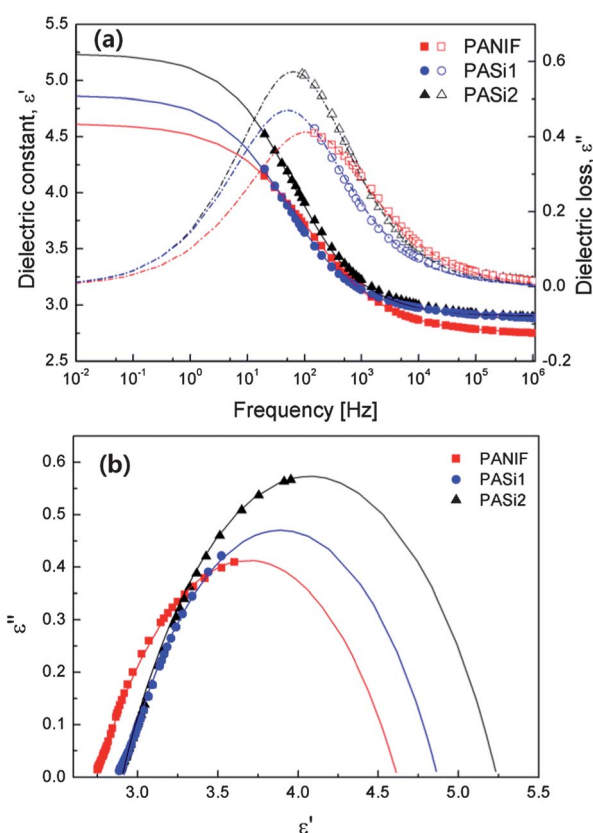


Fig. 8 (a) Dielectric spectra (ϵ' , closed; ϵ'' , open) and (b) Cole–Cole plot of the ER fluids containing pure PANIF, silica-modified PANIF, PASi1 and PASi2, respectively.¹³⁰

highest $\Delta\epsilon$ value shown by PASi2 indicated the highest ER effect because all samples exhibited a similar relaxation time.

3.5.1 Rare earth cations-doped TiO₂ and perovskites. TiO₂ is an attractive ER material owing to its high dielectric constant, and is a potential candidate for a high ER effect. On the other hand, in an initial study, only a low shear stress was discovered in anhydrous TiO₂-based ER fluids, which was attributed to its low conductivity or low loss factor.¹⁵⁶ The high dielectric constant correlated with the achievable polarizability and the appropriate dielectric loss was associated with the response time of polarization.^{149,150} The Zhao group developed TiO₂-based ER fluids with high yield stress by designing TiO₂ particles in special molecular and crystal structures for high polarizability and rapid polarization response.^{151–154}

A series of rare earth cation-doped TiO₂ were synthesized, the yield stress of which was more than 10 times larger than that of the pure TiO₂ ER fluid. Doping with large-radius rare earth ions

Table 1 Parameters in eqn (2) fitting to the dielectric spectra of the PANIF and silica-modified PANIF ER fluids¹³⁰

Sample	ϵ_0	ϵ_∞	$\Delta\epsilon$	λ	α
PANIF	4.62	2.75	1.87	0.0014	0.47
PASi1	4.87	2.91	1.96	0.0032	0.43
PASi2	5.24	2.90	2.34	0.0029	0.42

was performed to tune the dielectric and polarization properties of TiO_2 and achieve an enhanced ER effect. Light rare-earth doped anatase TiO_2 with different rare earth elements (lanthanum (La), cerium (Ce), yttrium (Y), *etc.*) and contents were prepared using a sol-gel method: the rare earth nitrate or chloride dissolved in distilled water was mixed with tetrabutyl titanate in anhydrous alcohol with the pH adjusted to 3–4 by HNO_3 and kept at room temperature for 12–48 h. The solid gel obtained was ground, heat treated in a furnace at 550°C before being used as an ER material. This process was also suitable for preparing Y-doped BaTiO_3 with a cubic crystal phase, which shows a larger ER effect than pure BaTiO_3 .^{155,156}

In view of the significance of interfacial polarization to the ER effect, particles with a large surface area were designed based on rare-earth-doped TiO_2 . The block copolymer (Pluronic F-127, $(\text{PEO})_{106}(\text{PPO})_{70}(\text{PEO})_{106}$)^{153,157} or long-chain surfactants (octadecylamine and poly(ethylene glycol))¹⁵⁴ was used as a template to direct the mesoporous structure of Ce-doped TiO_2 . The mesoporous structures in the calcined Ce-doped TiO_2 were confirmed by N_2 adsorption-desorption isotherms and TEM, as shown in Fig. 9. The BET surface area of the mesoporous particles was $110\text{ m}^2\text{ g}^{-1}$, which was much higher than that of Ce-doped particles without mesopores. Dielectric analysis of the Ce-doped mesoporous TiO_2 and pure TiO_2 was also carried out. In Fig. 10(a), the Ce-doped TiO_2 with mesopores showed the largest dielectric constant (or

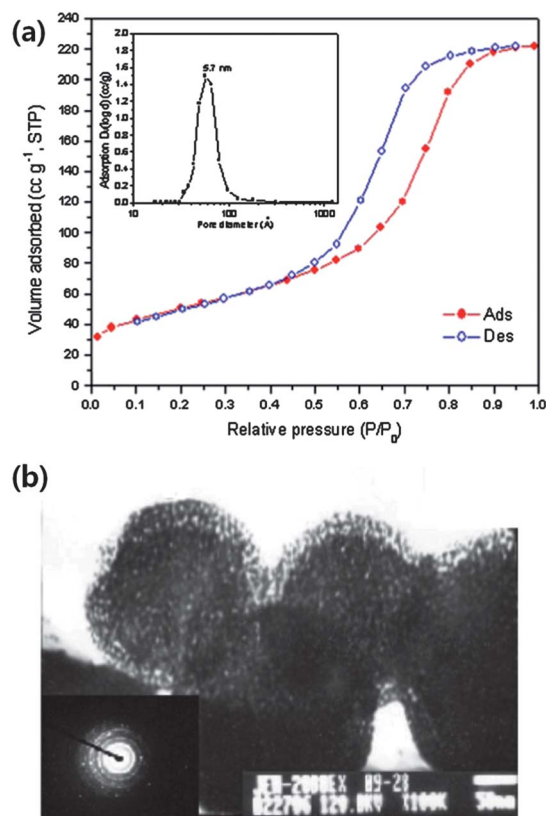


Fig. 9 N_2 adsorption-desorption isotherms of mesoporous Ce-doped TiO_2 (Ce/Ti = 7 mol%) with the corresponding pore size distribution (inset);¹⁵³ (b) TEM image of mesoporous Ce-doped TiO_2 (Ce/Ti = 8.5 mol%) and the electron diffraction pattern (inset).¹⁵⁴

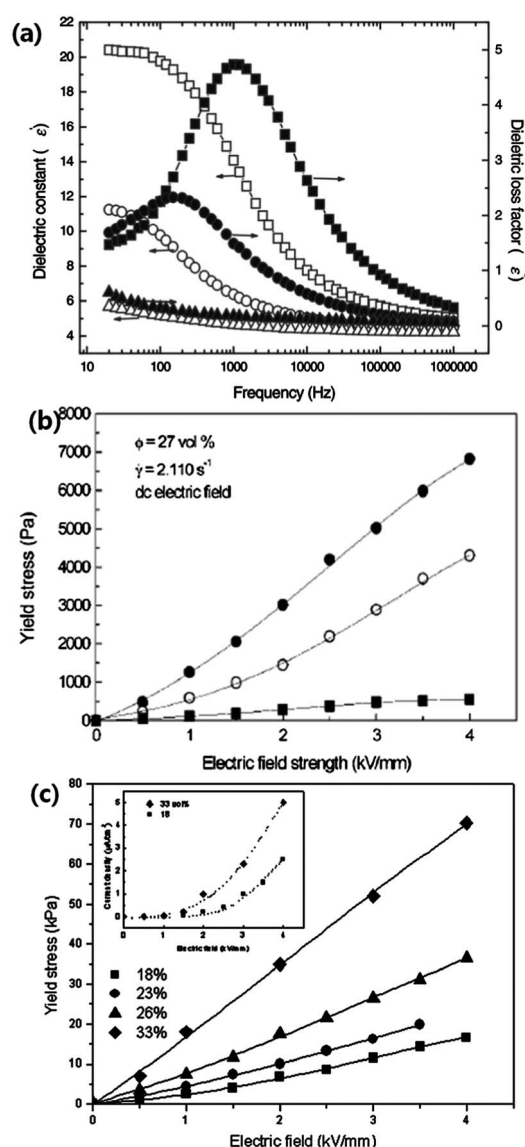


Fig. 10 (a) Dielectric spectra (dielectric constant: open; dielectric loss: closed) of pure nonporous TiO_2 (triangle), Ce-doped nonporous TiO_2 (circle) and Ce-doped mesoporous TiO_2 (square) ER fluids ($T = 25^\circ\text{C}$, particle volume fraction = 18%);¹⁵³ (b) static yield stress of pure nonporous TiO_2 (square), La-(open circle) and Ce-doped (closed circle) nonporous TiO_2 ($T = 25^\circ\text{C}$);¹⁵² (c) static yield stress of Ce-doped mesoporous TiO_2 ER fluids with different particle volume fractions and the corresponding current density (inset) ($T = 25^\circ\text{C}$, shear rate = 0.2 s^{-1}).¹⁵³

dielectric difference $\Delta\epsilon' = \epsilon'_0 - \epsilon'_\infty = 15$ compared to 7 and 2 for Ce-doped nonporous and pure TiO_2 , respectively). In addition, the Ce-doped mesoporous TiO_2 ER fluid also showed the shortest relaxation time with the maximum dielectric loss at the highest frequency. Both the largest dielectric difference and fastest dielectric response predicted a strong ER effect in the Ce-doped mesoporous TiO_2 . The same result was obtained and confirmed in Fig. 10(b) and (c), where at a fixed electric field strength ($E = 3\text{ kV mm}^{-1}$), the Ce-doped TiO_2 ER fluid (taking the 26% sample (represented by the closed triangle) as a reference) showed the highest yield stress of $\sim 25\text{ kPa}$ compared to 0.5 kPa and 0.3 kPa , as shown by the Ce-doped nonporous

TiO₂ and pure TiO₂ ER fluids. As a result, the mesoporous structure in the ER particles could contribute to the ER effect, which was also confirmed using the mesoporous metal–organic framework of Cu₃(BTC)₂ in an ER fluid.¹⁵⁸

3.5.2 Nanostructured titanate or TiO₂. A strong ER effect was achieved in ER fluids containing inorganic materials including the aforementioned rare earth-doped TiO₂. On the other hand, micron or sub-micron inorganic particles always resulted in poor settling stability due to the large density mismatch between the particles and suspending liquid, except for mesoporous particles, which require a high cost and complicated synthesis process.¹⁵⁴ ER particles in the nanoscale range could solve the sedimentation problem but also resulted in a weaker ER effect. TiO₂ with designed nanostructures ranging from general nanospheres to nanotubes and sea-urchin-like hierarchical particles were synthesized. One-dimensional nanostructures, such as nanowires and nanotubes, were then introduced to design the ER materials with better stability and a relatively strong ER effect. Titanate nanowhiskers (Fig. 11(a))^{159,160} were synthesized from TiO₂ nanopowders using a hydrothermal procedure, but they exhibited considerably higher yield stress than the original nanopowder (Fig. 12(a)). Fig. 12(b) shows the sedimentation ratio of the ER fluids containing TiO₂ nanoparticles and microparticles compared to the ER fluid of TiO₂ nanowhiskers. The sedimentation ratio was defined simply as follows:

$$\text{Sedimentation ratio} = 1 - \frac{\text{volume of the supernatant liquid}}{\text{volume of the entire suspension}} \quad (3)$$

in which the volume of the supernatant liquid was calculated from the measured height as a function of time. Sedimentation in the ER fluid of the TiO₂ microparticles appeared directly and was so fast that the particles settled almost completely after 10 days (~240 h). In the ER fluid of TiO₂ nanoparticles, sedimentation was detected after 5 days. On the other hand, there was no obvious sedimentation observed in the TiO₂ nanowhiskey ER fluid.

A kind of sol–gel-synthesized TiO₂ nanoparticle in the presence of acetic acid was reported to express an ER effect. Owing to the polar molecule (–OH) dominating on the TiO₂ particles, the

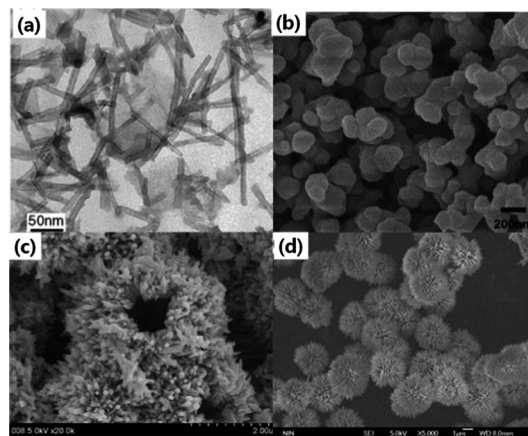


Fig. 11 (a) TEM image of titanate nanowhiskers and (b) SEM images of TiO₂ nanoparticles, (c and d) sea-urchin like TiO₂ spheres.^{159,161–163}

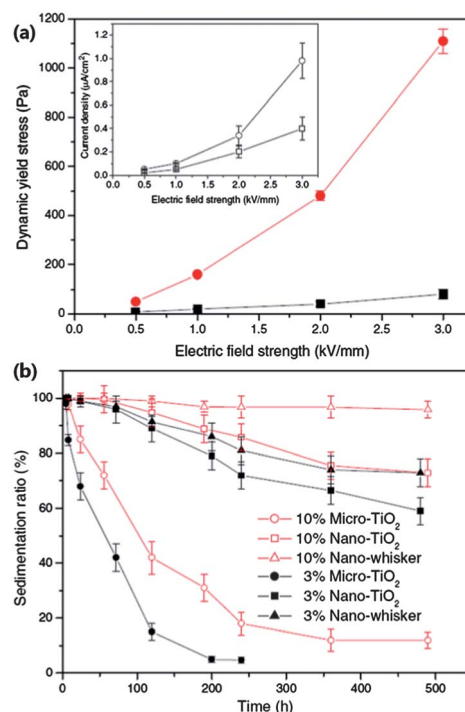


Fig. 12 (a) Dynamic yield stress vs. electric field strength plots of ER fluids containing tube-like TiO₂ nano-whiskers (circle) and raw materials of TiO₂ nanoparticles (square) (inset: corresponding current density of the ER fluid); (b) sedimentation ratio of ER fluids containing tube-like nano-whiskers (triangle), raw materials of TiO₂ nano-particles (square), and TiO₂ microparticles (circle) with the same volume fraction of 3% (solid) and 10% (open).¹⁶⁰

static yield stress can be up to 130 kPa in an electric field of 4 kV mm^{−1}.¹⁶² In addition, Cheng *et al.*¹⁶¹ reported an ER fluid containing sea-urchin-like TiO₂ hollow spheres (Fig. 11(c)), indicating its advanced ER effect compared to pure TiO₂ particles with a smooth surface.

To combine the positive influence of doped rare earth cations and the morphology on the nanoscale, monodisperse Cr-doped sea-urchin-like hierarchical TiO₂ particles (Fig. 11(d)) containing 1D nanorods on the surface of TiO₂ core spheres were synthesized by a solvothermal process rather than a common sol–gel-hydrothermal method. Compared to the Cr-doped smooth non-hierarchical TiO₂ particles, the novel particles exhibited higher static yield stress and storage modulus owing to their thicker column-like structures impacted by interparticle friction and mechanical cohesion under shear.

Overall, for TiO₂-based ER materials with a high dielectric constant, doping with rare earth ions on the nanoscale will be helpful for producing ER materials with a high ER effect and good settling stability.

3.6 Giant ER materials

Urea-coated BaTi(C₂O₄)₂ nanoparticles were the first GER materials prepared using a co-precipitation method, in which barium chloride in distilled water and oxalic acid in water with titanium tetrachloride added drop wise were mixed before adding

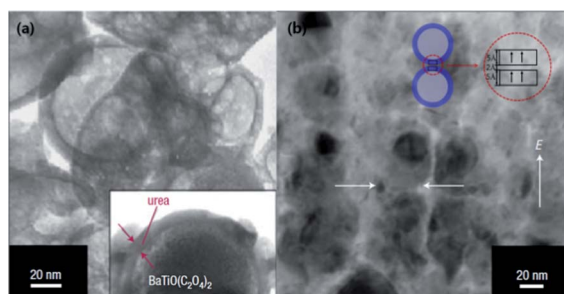


Fig. 13 (a) TEM images of urea coated $\text{BaTi}(\text{C}_2\text{O}_4)_2$ nanoparticles; (b) optical microscopy image of the particle chains fixed by epoxy under an applied electric field of 2 kV mm^{-1} . The inset shows a schematic diagram of the surface polarization layer model: the gap between the two flattened layers is $2 \text{ \AA}$ and the thickness of the saturation polarization layer is $5 \text{ \AA}$.⁴⁶

urea. The TEM image in Fig. 13(a) shows the coated $\text{BaTi}(\text{C}_2\text{O}_4)_2$ nanoparticles by soft urea molecules with a thickness of $\sim 3\text{--}10 \text{ nm}$. Urea with a large molecular dipole moment was considered to be helpful to the interparticle forces due to the formation of an aligned dipolar layer in the contact region between the two coated particles (Fig. 13(b))⁴⁶ where there was a surface saturation polarization. Therefore, there was also a particle-size-effect on the ER performance because of the difference in surface area: doping with Rb reduced the size of the $\text{BaTi}(\text{C}_2\text{O}_4)_2$ particles coated with urea but achieved double the yield stress.¹⁶⁴

Later, polar molecule-modification of SiO_2 and TiO_2 nanoparticles was applied to enhance the ER performance of these particles. In addition to the first applied polar molecule, urea,^{165–168} other polar molecules, such as triethanolamine^{169,170} and acetamide,¹⁷¹ were attempted as modification agents to inorganic particles. On the other hand, only the *in situ* added polar molecules could generate a GER effect rather than post modification of the synthesized inorganic particles. From the preparation of GER fluids, a high particle weight (or volume) fraction (nearly 50 wt% or even higher) was noticed. Therefore, it is essential to use suspension oils with low viscosity to reduce the off-state viscosity of the fluid. The wettability of the dispersed particles was also considered, upon which hydrophilic groups (hydroxyl)-terminated silicone oil was examined as a suspending liquid in the GER system.¹⁷² On the other hand, surface-modified TiO_2 nanoparticles with SDS or SDBS prepared by *in situ* hydrolyzation exhibited excellent wettability to silicone oil and good dielectric properties.^{173,174}

To compare the various ER materials, Table 2 summarizes the yield stresses of their corresponding ER fluids at an applied electric field strength of 2 kV mm^{-1} . Suitable applications are also listed in the table according to the literature.¹⁷⁵ A commercial ER fluid (Rheoil system) was included in many investigations on the application of ER fluids based on commercial ER fluids. In addition, application investigations, such as ER finishing, were carried out on starch-^{176,177} or liquid-crystal-based ER fluids.¹⁷⁸ A high yield stress is required for engineering devices.¹⁷⁹ On the other hand, ER fluids generating high yield stress always have other problems, for example, serious sedimentation in an inorganics-based ER fluid and high zero field viscosity of a GER fluid. Cooperation should be closely established between researchers in the fields of ER materials and applications to fully utilize the materials mentioned in this chapter.

4. Electrorheological characteristics

A variety of characterization methods are needed to understand the rheological performance of an ER fluid in more detail. The ER properties of ER fluids are observed mostly in a steady shear or oscillation flow using a rotational rheometer, but can also be detected in compression flow using specially designed devices. Among the many characteristics of ER fluids, the rheological properties including yield stress, shear stress and viscosity, dynamic modulus (storage modulus, G' and loss modulus, G''), and complex compliance under a range of test modes have been studied extensively.

On the other hand, one significant examination before the rheological measurement is to observe the particulate aggregations in an ER fluid by an optical microscopy (OM). Movement and aligned structures of the particles differ according to the response of the particles to the electric field, and the volume fraction of the particles, which can be observed and confirmed by OM. Fig. 14 shows a schematic diagram and the real performance of the snowman-like PMMA–PANI particles before and after applying an electric field.¹¹⁸ The randomly dispersed particles transform to aligned particle-chains with connected particles in a head-to-head or head-to-tail form rather than particle-clusters when using spherical ER particles.¹²⁶ A similar phenomenon was discovered by Shin *et al.*¹⁸² using the PANI-coated symmetric dimer particles.

In a polyhedral silsesquioxane (POSS)-based polydimethylsiloxane (PDMS) suspension with PS fillers,¹⁸³ the ER activity was generated from the polar groups, such as trisulfonic acid ethyl

Table 2 Various ER fluids and their corresponding characteristics

	Conducting polymers and their composites	High dielectric inorganics	Rheoil 3.0 ^a	Giant ER materials
Particle size (μm)	1–10	0.1–1	1–10	0.01–0.1
Concentration (vol%)	5–20	5–25	~ 50 ^b	~ 30 ^c
Yield stress (Pa) at 2 kV mm^{-1}	10^2 to 10^3	10^3	10^3	10^4
Potential applications	Tactile element, ¹⁸ shock absorber ¹⁸⁰	Tactile element ¹⁶⁵	Tactile element, ¹⁸¹ damper	Droplet sensing, microfluidic chip and pump ⁵⁰

^a A commercial ER fluid of Fluidicon® (Germany) basing on polyurethane microparticles. ^b Estimated from the density of the ER fluid (1.04 g cm^{-3}).

^c The giant ER fluids containing 10 g particles in 3 ml oil is defined as 30 vol%.

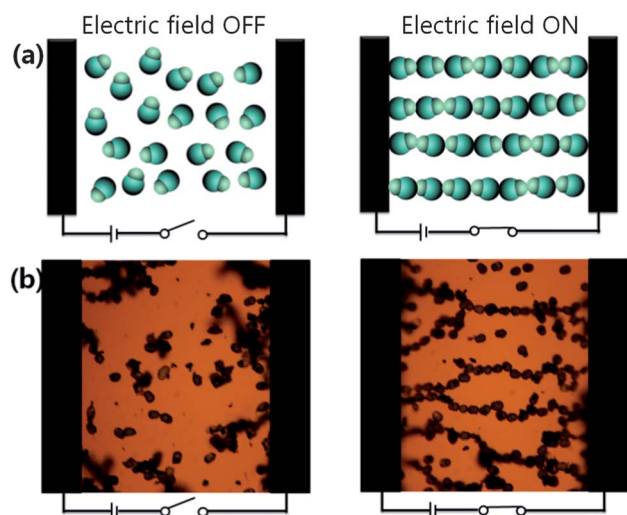


Fig. 14 Schematic diagram (a) and OM (b) images of the dispersed particles (snowman-like PMMA with PANI coating) of an ER fluid injected between the gap of two electrodes ($\sim 350 \mu\text{m}$) before and after applying an electric field (300 V).¹¹⁸



Fig. 15 OM image of PS spheres after removal from the TSAiB POSS/PS/PDMS suspension following ER testing.¹⁸³

and tris sulfonic acid isobutyl (TSAiB) in the POSS molecule. The POSS particles were observed to coat the PS particles and induce the aggregation of PS particles by the application of an electric field (Fig. 15). This is why the ER fluid showed an order of magnitude higher ER activity despite the concentration of sulfonated POSS being as low as $\sim 1 \text{ wt}\%$.

In addition to direct observations by the OM, von Pfeil *et al.*¹⁸⁴ examined a two-fluid continuum model of particle structures in the quiescent suspension and developed shear flow using a conservation equation for the particle flux obtained from the momentum balance.

$$\frac{\partial \phi}{\partial t} + \langle u \rangle \cdot \nabla \phi = -\nabla \cdot j \quad (4)$$

where ϕ is the particle volume fraction, $\langle u \rangle$ is the suspension average velocity, $j \equiv \phi(U - \langle u \rangle)$ is the particle flux and U is the local mean velocity of the particulate phase. By introducing the stress from the particles in the mass balance, column and stripe formation occurred in an electric field in the absence of flow and under shear flow, respectively, as a result of a structure-related decrease in free energy of polarization and the flow-stabilized

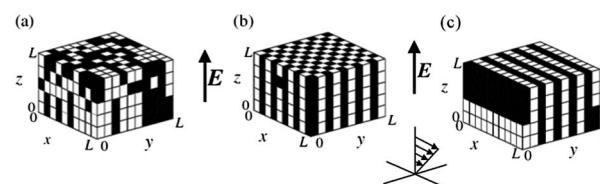


Fig. 16 Initial structure (a) and structures after the application of an electric field to a quiescent (b) and flowing (c) suspension. The dark and light cubes represent the positive and negative fluctuations, respectively.¹⁸⁴

fluctuation under shear. Fig. 16 shows the evolution of the concentration fluctuations.

As observed by confocal microscopy, the application of an electric field was also found to be vital for controlling the self-assembly of colloidal particles and creating newly structured crystalline lattices,^{185,186} in addition to the similar effect induced by oscillation shear flow.¹⁸⁷ In a colloidal suspension of either hard or soft silica dumbbells, the random plastic solids were transformed to fully ordered crystals by the application of an electric field due to dielectric mismatch between the particles and medium. Different crystal structures were obtained by varying the particle aspect ratio and electric field strength.¹⁸⁵ An electric field induced phase transition was also discovered in concentrated suspensions of colloidal rods (fd-virus particles) according to the interactions between the polarized electric double layer by varying the strength and frequency of the ac field.^{188,189} All rheological performances of the colloidal suspensions are closely related to the inner particulate arrangements or crystal structures. Therefore, examining the particulate structures theoretically and in real space is important for understanding the rheological properties of an ER fluid in either flow or electric field conditions, or in both fields measured in the modes discussed below. However, when the field is along the flow direction, the formed particle chains or crystals are in the direction of flow, generating reduced viscosity. It has been recently applied in extracting and transporting crude oil,¹⁹⁰ and reducing blood viscosity by the application of a magnetic field.¹⁹¹

4.1 Flow curve analysis

Steady shear flow in an ER fluid can be constructed in a range of modes including the shear rate sweep (controlled shear rate, CSR), constant shear rate, and shear stress sweep (controlled shear stress, CSS) using either a concentric cylinder or parallel-plate cell. They are useful for observing both the dynamic and static yield stresses of the ER fluid as well as their dependence on the electric field strength. The on-off switch properties in an ER fluid are also indicated by the steady shear flow measurements at a fixed shear rate. The shear rate sweep is a commonly applied method of measuring the flow curves and dynamic yield stresses of an ER fluid, from which the stability of the formed particle structures can be predicted. To analyze the flow curves of an ER fluid, the simple Bingham model is generally used by introducing the dynamic yield stress (τ_y) and shear viscosity (η) as follows:

$$\tau = \tau_y + \eta \dot{\gamma} \quad (5)$$

The dynamic yield stress can be obtained by extrapolating the flow curve to zero shear rates. Fig. 17(a) shows a re-plot of the data using the data in ref. 192, the flow curves of an ER fluid containing nitrogen-enriched carbonaceous PANI nanotubes (15 vol%) heated to 600 °C (N-CT-600), which were fitted by eqn (5). On the other hand, in a dilute ER fluid or an ER fluid with a lower ER effect, the particle chains formed can be destroyed easily but are difficult to reform immediately under shear flow. A decrease in shear stress was observed until a critical shear rate when the chain structures in the ER fluid were totally broken. Fig. 17(b) shows the flow curves of a 10 vol% ER fluid based on the silica nanoparticles-modified PANI nanofiber¹³² under CSR mode. Without an electric field, the fluid exhibited Newtonian viscous flow by showing a linearly increased shear stress by increasing the shear rate. When an electric field was applied, yield stresses appeared and increased with increasing applied electric field strength. On the other hand, the shear stress at each electric field decreased with increasing rate, showing a minimum value at a critical shear rate. A modified rheology equation of state, called the Cho–Choi–Jhon (CCJ) model, was suggested to model this type of flow curve.

$$\tau = \frac{\tau_y}{1 + (t_2\dot{\gamma})^\alpha} + \eta_\infty \left(1 + \frac{1}{(t_3\dot{\gamma})^\beta} \right) \dot{\gamma} \quad (6)$$

where η_∞ is the viscosity at infinite shear rate and is considered to be the viscosity in the absence of an electric field; t_2 and t_3 are the time constants used to describe the change in shear stress; the exponent α is related to the decrease in shear stress at a low shear rate region, whereas β in the range of 0 to 1.0 is for the high shear

rate region. Using the optimal parameters, the CCJ model shows a much better fit of the flow curves than the Bingham model. Table 3 lists the parameters applied in both models.

In addition to the rheological models mentioned above, Seo *et al.*¹⁹³ recently introduced a new model based on the Papanastasiou model¹⁹⁴ to analyze the flow behavior of ER fluids.

$$\tau = \tau_y \left(1 - \frac{(1 - \exp(-a\dot{\gamma}))}{(1 + (a\dot{\gamma})^\alpha)} \right) + \eta\dot{\gamma} \quad (7)$$

where a is the time constant defined as the reciprocal of the critical shear rate. A power-law form in the first term of eqn (7) was used for the shear rate dependence to account for the break-up and reformation of the aligned particle-chain structure. The fitting results in the study indicated the superiority of the CCJ model in analyzing flow curves. Depending on the parameters, eqn (7) can describe both Newtonian ($\tau_y = 0$) and Bingham fluids ($\alpha = 0$, $a = 0$) as well as other rheological models.¹⁹⁵

ER fluids with a high volume fraction (near or higher than 50 vol%) of particles are always measured using a parallel-plate cell owing to their high viscosity. Fig. 18 shows the flow curves of a GER fluid based on urea-coated Ba_{0.8}(Rb)_{0.4}TiO(C₂O₄)₂ nanoparticles prepared using a co-precipitation procedure.¹⁶⁴ In contrast to the flow curves in Fig. 17, the shear stress in Fig. 18(a) first increases with increasing shear rate, reaching a maximum, and then decreasing with further increases in shear rate. The inset image in Fig. 18(a) shows that, the GER fluid sample is expelled from the loading region between the two plates as the shear rate is increased, resulting in a decrease in shear stress. To confine the loaded samples, Orellana *et al.*¹⁹⁶ used a non-conducting cylindrical ring (installed as shown in Fig. 18(b)) and applied a fixed normal force when a GER fluid containing strontium titanyl oxalate particles was measured using a parallel-plate geometry. The shear stress under each condition (confined or free edge, normal force controlled) showed a steady value over the entire shear rate range. From this point of view, advanced and flexible

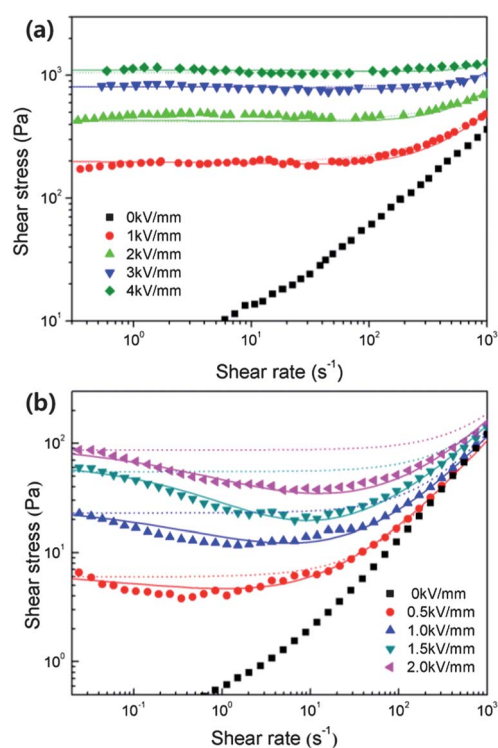


Fig. 17 Flow curves of N-CT-600 ER fluid (15 vol%)¹⁹² (a) and PASi2 ER fluid (10 vol%)¹³⁰ (b). The dot and solid lines in each plot were fitted by the Bingham and Cho–Choi–Jhon (CCJ) model, respectively.

Table 3 Fitting parameters of the Bingham and CCJ models to the flow curves of N-CT-600 and PASi2 ER fluids

Sample	Model	Parameters	Electric field strength kV ⁻¹ mm ⁻¹			
			1	2	3	4
N-CT-600	Bingham	τ_y	195	420	800	1050
		η	0.3	0.3	0.2	0.2
	CCJ	τ_y	200	430	820	1100
		t_2	0.001	0.001	0.001	0.001
		α	0.6	0.8	0.8	0.9
		η_∞	0.2	0.3	0.4	0.4
		t_3	0.01	0.01	0.01	0.002
		β	0.01	0.1	0.2	0.3
	Bingham	τ_y	0.5	1	1.5	2
		η	6	23	55	87
PASI2	Bingham	τ_y	0.11	0.10	0.10	0.10
		η	8	28	65	98
	CCJ	τ_y	1.5	2.3	2.5	2.6
		t_2	0.25	0.40	0.60	0.45
		α	0.08	0.08	0.10	0.10
		η_∞	0.01	0.004	0.0025	0.002
		t_3	0.50	0.60	0.65	0.70
		β				

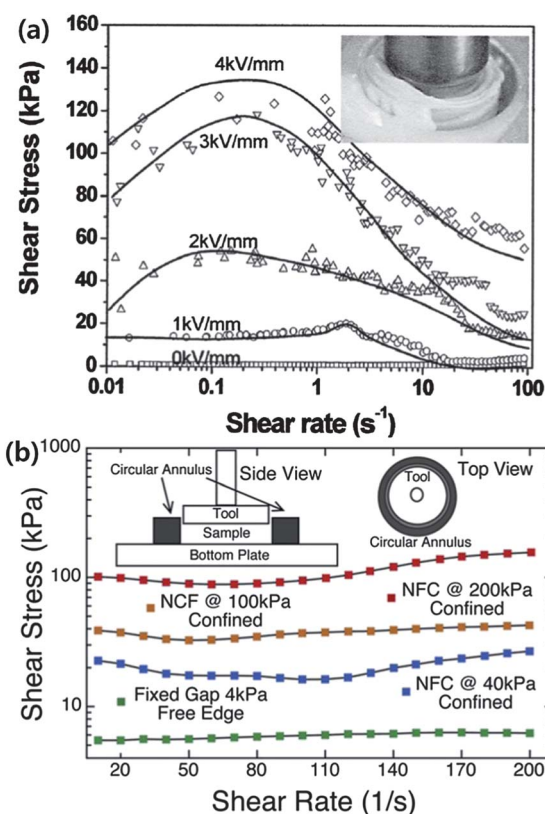


Fig. 18 (a) Shear stress vs. shear rate of urea-coated $\text{Ba}_{0.8}(\text{Rb})_{0.4}\text{-TiO}(\text{C}_2\text{O}_4)_2$ GER fluid sample measured using a parallel-plate geometry under different electric field strengths;¹⁶⁴ (b) shear stress vs. shear rate of STO-based GER fluid measured using a parallel-plate cell of a commercial rheometer MCR 301 at 5 kV mm^{-1} for different boundary conditions and applied loads. Inset: schematic diagram of the annulus used to confine the sample.¹⁹⁶

testing geometries will be needed to understand the ER behavior better and tailor the fluids to specific applications.

4.2 Yield stress and dynamic moduli

4.2.1 Dynamic and static yield stress. From the measured flow curves, the dynamic yield stress was determined by extrapolating the shear stress to a zero shear rate. On the other hand, the static yield stress could be obtained in the CSS measurement by observing the viscosity or strain as a function of the shear stress. In general, the correlation between the yield stress and electric field strength (E_0) can be represented by the following:

$$\tau_y \propto E_0^\alpha \quad (8)$$

The power law index α was suggested to be 2.0 in the polarization model.¹⁹⁷ Further studies indicated that α always deviated from 2.0 according to the materials of the particles and their concentration, shape and electric field strength.^{198–200} In particular, for conducting ER particles at high electric field strength, the ER effect was considered to have been caused by a conductivity mismatch between the particles and liquid media rather than a dielectric constant mismatch. The power law dependence α was found to approach 1.5 for these ER fluids. Many studies

reported different values of α in the range of 1.5–2.0 or even smaller than 1.5.

Recently, the critical electric field strength, E_c , was introduced. E_c was defined as the intersection of the fitted power law lines with two exponents, 2.0 and 1.5 in the low and high electric field region, respectively. The simple hybrid yield stress equation for a broad electric field strength range can be expressed as follows:

$$\tau_y(E_0) = \alpha E_0^2 \left(\frac{\tanh \sqrt{E_0/E_c}}{\sqrt{E_0/E_c}} \right) \quad (9)$$

where the parameter α depends on the dielectric properties of the fluid, the particle volume fraction, and the critical electric field. At both low and high limiting electric field strengths, eqn (9) can be changed to

$$\tau_y = \alpha E_0^2 \propto E_0^2 \text{ for } E_0 \ll E_c \quad (10a)$$

$$\tau_y = \alpha E_0^2 \sqrt{E_0/E_c} \propto E_0^{1.5}, \text{ for } E_0 \gg E_c \quad (10b)$$

A universal scaling equation was then introduced using eqn (9), to place the yield stress data into a single curve.

$$\hat{\tau} = 1.313 \hat{E}^{3/2} \tanh \sqrt{\hat{E}} \quad (11)$$

where $\hat{E} \equiv E_0/E_c$, $\hat{\tau} \equiv \tau_y(E_0)/\tau_y(E_c)$, and $\tau_y(E_c) = 0.762\alpha E_c^2$.

Fig. 19(a) shows the dynamic yield stress of two ER fluids based on PANI fiber and PASi, as a function of the electric field strength. E_c of each ER fluid was 1.25 and 1.82 kV mm^{-1} for the

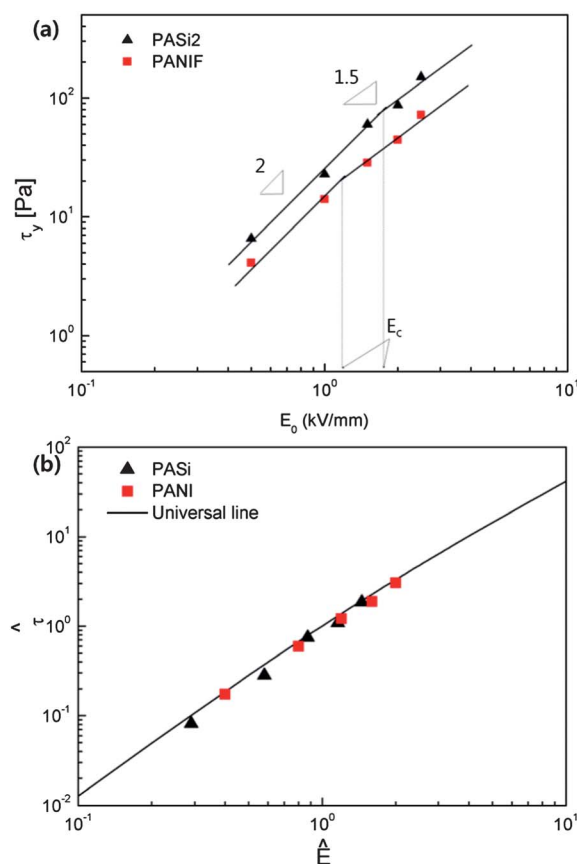


Fig. 19 (a) Dynamic yield stress vs. electric field strength and (b) universal curve for the PANI fiber ER fluid and PASi ER fluid.¹³⁰

PANI fiber and PASi, respectively. The original data (τ_y , E_0) in Fig. 19(a) was replaced with $(\hat{\tau}, \hat{E})$ using the E_c values and replotted (Fig. 19(b)). The solid line in Fig. 19(b) generated from eqn (11) collapsed all the yield stress points in a universal line, indicating the significance of eqn (11) in constructing the yield stress master curve for ER fluids.

On the other hand, the yield stress of the GER fluid showed an almost linear dependence on the applied electric field, which could not be scaled by eqn (11). Therefore, a new equation with diverse limiting dependence on the electric field strength was introduced.²⁰¹

$$\tau_y(E_0) = \left(\frac{\alpha E_0^2 I_1(E_0/E_c)}{(E_0/E_c) I_0(E_0/E_c)} \right) \quad (12)$$

where I_1 and I_0 are the modified Bessel functions with an order of 1 and 0, respectively. The limiting properties of eqn (12) were examined as follows: when $E_0 \ll E_c$, $I_1(E_0/E_c) \propto E_0$, and $I_0(E_0/E_c) = \text{constant}$,

$$\tau_y(E_0) \propto E_0^2$$

when $E_0 \gg E_c$, $I_1(E_0/E_c) \propto E_0$ and $I_0(E_0/E_c) \propto e^{E_0/\sqrt{2\pi E_c}}$,

$$\tau_y(E_0) \propto E_0$$

Therefore, another scaling equation was proposed using $\tau_y(E_c)$ and is defined with $\hat{\tau}$ and $\hat{E}$ as follows:

$$\hat{\tau} = \frac{2.242 \hat{E} I_1(\hat{E})}{I_0(\hat{E})} \quad (13)$$

in which $\tau_y(E_c) = 0.446 \alpha E_c^2$.

The static yield stress data referring to two GER fluids, sample 1 from ref. 46 and SiO₂ (2.3 wt%)-CTP from ref. 202, was replotted in the inset in Fig. 20 and rearranged using eqn (13). The rearranged data were all collapsed by eqn (13) in a single curve using the critical electric field strength for each GER fluid.

Although the proposed correlations of eqn (11) and (13) are not derived from first principles, they are important for representing the data by experimentalists. The beauty of these

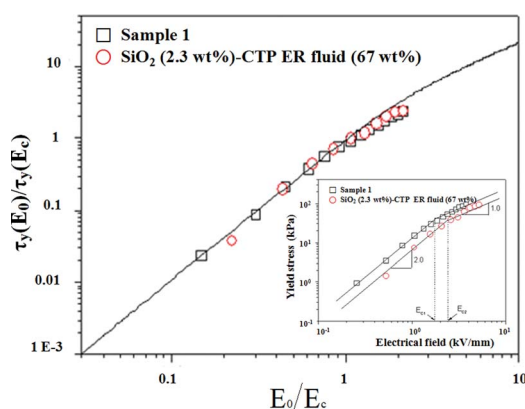


Fig. 20 The yield stress vs. electric field strength for two GER fluids. Inset: the static yield stress vs. electric field strength for the two GER fluids.²⁰¹

correlations is that most of the ER/MR experimental data can be collapsed with a few parameters,²⁰³ suggesting a guideline for better ER performance.

4.2.2 Dynamic oscillation. The dynamic oscillation test is a general way of determining the nonlinear viscoelastic properties of an ER fluid to detect the formation of chain structures. This is constructed as a function of frequency at a constant strain amplitude selected in the linear viscoelastic region. On the other hand, Yin *et al.*¹²⁹ suggested a novel oscillation measurement from a study of TiO₂ coated PANI nanocable ER fluids: controlled shear stress applied during the oscillation test. Fig. 21(a) shows the frequency dependence of the storage modulus (G') and loss modulus (G'') of the TiO₂-coated PANI nanocable ER fluid at a stress of 20 Pa in the linear viscoelastic region. The ER fluid behaved like a solid material, showing a higher G' than G'' . All the G' values measured at different electric field strengths remained relatively constant over a broad frequency range, and increased with increasing electric field strength.

In this special study of TiO₂-coated PANI nanocables as ER materials, the storage and loss moduli were also measured as a function of the shear stress at a constant frequency. As shown in Fig. 21(b), both G' and G'' reached a plateau as the shear stress was increased, and then decreased abruptly when the shear stress reached the yield point. The sharp decrease in viscoelastic moduli indicated rupture of the structure of the particle chains in the ER fluid. This can be considered an effective way of obtaining the yield stress of an ER fluid in oscillatory measurements.

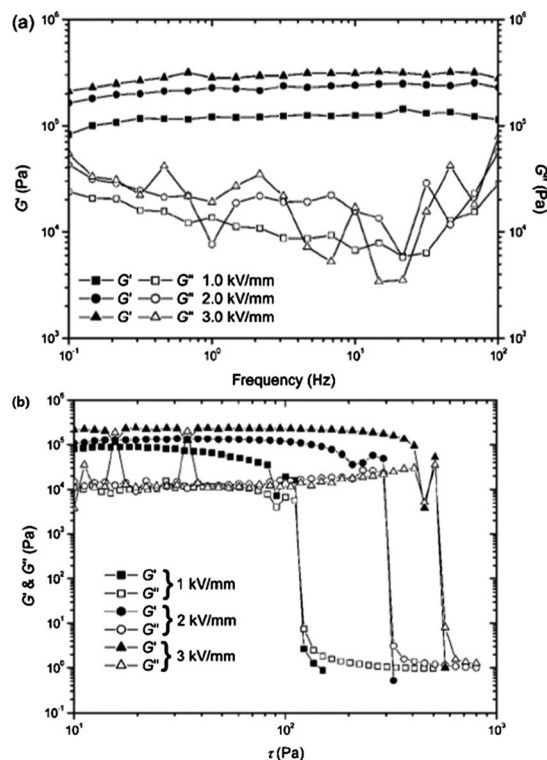


Fig. 21 The frequency (a) and shear stress (b) dependence of G' and G'' of the ER fluid containing TiO₂ coated non-conducting PANI nanocables.¹²⁹

In contrast to the rotational rheometer, a vertical oscillation rheometer (VOR)²⁰⁴ was designed for the dynamic oscillation test, the mobile cylinder cell of which moved vertically to contact the ER fluid contained in the immobile cup cell. The dynamic moduli (G' and G'') can be calculated from the output data on an oscilloscope including the phase angle (δ), maximum amplitude of strain (z_0) and force (F_0).

$$G' = \frac{F_0 \cos \delta \ln a}{2\pi/z_0}, G'' = \frac{F_0 \sin \delta \ln a}{2\pi/z_0} \quad (14)$$

where a is the ratio of the diameters of the outer cup and inner cylinder, and l is the height of the inner cylinder. The storage modulus obtained from the VOR provided similar values to those obtained from the commercial rotational rheometer over the entire range of frequencies at the same electric field strength. The loss modulus shown by the VOR was slightly lower than that obtained from the commercial rotational rheometer.²⁰⁵

On the other hand, the experimental data for dynamic response of ER fluids was limited in the small strain amplitudes for linear viscoelastic behavior. Therefore, the dynamic behavior of ER fluids at large strain amplitude was simulated in the particle level that agreed with the experiments.²⁰⁶ For the large strain amplitude, the ER response was viscoplastic with the complex viscosity ($|\eta^*|$) depending only on the dimensionless oscillation shear rate amplitude, $\tilde{\omega}\gamma_0$, which differed from the linear regime with a dependence on $\tilde{\omega}$. A Pipkin diagram was constructed to summarize the qualitative dependence of the flow behavior in laminar oscillatory shear flow. As shown in Fig. 22, the flow regimes were mapped as functions of the dimensionless frequency ($\tilde{\omega}$) and strain amplitude (γ_0).

4.3 Compressive stress

In addition to studies of ER materials for a high ER effect, there are other ways to enhance the stress performance of ER fluids. The compressive and tensile stresses of the ER fluids were much higher than the shear stress under the same applied electric field strength. The Tian group carried out considerable work on this topic by measuring a zeolite ER fluid under compression or elongation using the measurement system shown in Fig. 23.^{207–209}

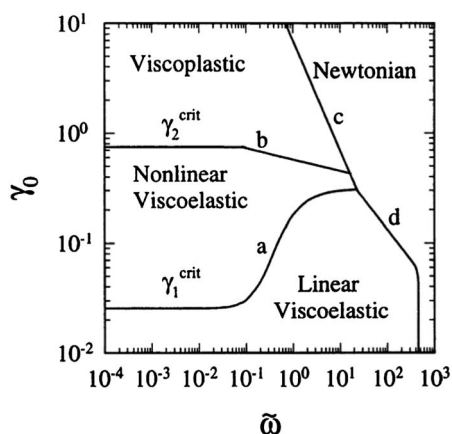


Fig. 22 The Pipkin diagram of the dynamic rheological behavior of ER fluids.²⁰⁶

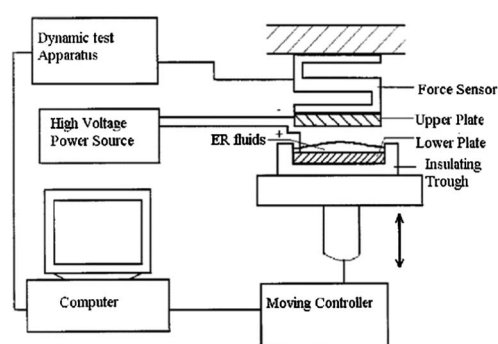


Fig. 23 Schematic diagram of the elongation and compression test system.²⁰⁷

In the compression test along the direction of the electric field, the compressive stress obtained represents the resistance of the ER fluid. Compared to the shear yield stress (measured by the HAAKE RV20 rheometer) and tensile yield stress in elongation, the compressive stress taken from the end point of the compression tested showed the highest value (Fig. 24). A nominal shear yield stress (τ_E) was suggested by transforming the compressive flow to shear flow using the following equation:

$$P = \tau_E D/3h \quad (15)$$

where P represents the compressive stress; and D and h are the diameter and gap distance of the parallel plate electrodes, respectively. Using eqn (15), an ultrahigh yield stress larger than 100 kPa was achieved in slow compressive flow under an electric field of $<0.6 \text{ kV mm}^{-1}$.²⁰⁸ Moreover, a pre-loading of compression after applying an electric field can improve the shear yield stress up to ten fold.^{209,210}

5. Concluding remarks

Based on the use of various ER electro-responsive soft materials, including mainly conducting polymers, dielectric inorganics with or without modification and composite materials with special structural characteristics (fibers, core-shell spherical or nonspherical particles, *etc.*), researchers in this field have made considerable progress in the development of new materials, characteristic methods and corresponding mechanisms. Various synthesis strategies, *e.g.* interfacial oxidation, microemulsion, Pickering emulsion, grafting for polymers, coprecipitation, hydrothermal methods, and a sol-gel procedure for inorganic particles, have been developed the production of ER materials with desired chemical and physical properties. Nevertheless, considerable work is needed to increase the practical applications of ER fluids. A thin layer coating with conducting material or polar molecules can result in rapid polarization and a strong ER effect but the thermostability, long-term settling stability and off-state viscosity should also meet the requirements of individual applications. Thereby, researches of electrorheology are still widely open for extensive studies regarding theoretical and experimental understandings of both basic mechanisms on its origin independent of different electroresponsive materials based on their electric and dielectric characteristics, and complex rheological flow behaviors. Researches of new GER fluids and

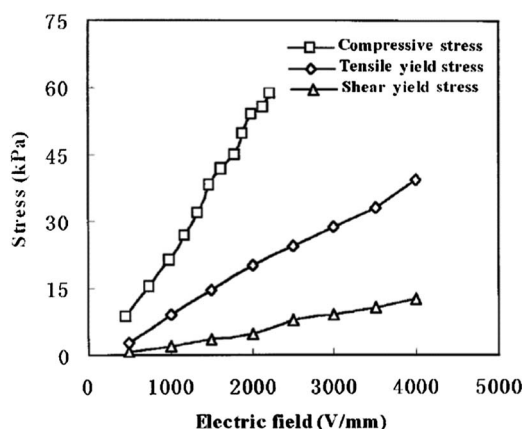


Fig. 24 Stress of the ER fluid measured in different flow motions.²⁰⁷

related areas of electric-field-induced transitions and crystal ordering are also important. On the other hand, for commercial applications, it is important to design a small, fine device with an equipped power supply but with high supply ability and high level safety. Once these are solved, it should be possible to commercialize ER fluids in a range of current and newly developed industries.

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